

Hsu, Chun-Yuan

 My Portfolio |  chyhsu@umich.edu |  Chyhsu |  Chun-Yuan Hsu

Skills

Languages: Python (Advanced), C++, Go

AI & ML: PyTorch, LLM Integration (AWS Bedrock), RAG (ChromaDB), Embeddings, Semantic Search

Cloud & DevOps: Docker, Kubernetes, AWS, GCP, GitLab CI/CD, Grafana, NATS

Databases: MongoDB, Couchbase, ChromaDB, PostgreSQL, SQLite

Frameworks/Systems: Node.js, React, Linux (Debian, Arch), Git, Scrum

Education

Master of Science in Data Science, University of Michigan – Ann Arbor, MI, USA Sep 2025–Present

Master of Science in Computer Science, National Tsing Hua University – Hsinchu, Taiwan Sep 2022–Jan 2025

Thesis: Quantum Event Identification and Learning Procedures

Bachelor of Science in Civil Engineering, National Cheng Kung University – Tainan, Taiwan Sep 2018–Jun 2022

Experience

Backend R&D Internship, QNAP, Taiwan Jan 2025 – Jul 2025

- Architected a semantic **Jira issue search engine** using AWS Bedrock LLM and ChromaDB embeddings, improving developer resolution productivity by 50%.
- Engineered a **Jira search MCP server**, enabling developers to query issue tracking directly via coding IDEs.
- Refactored Device Avatar microservices from Python to **Go**, achieving a 30% performance gain and optimizing deployment on Kubernetes.
- Diagnosed and patched a critical memory leak in cloud production by correlating **Grafana metrics** with execution traces.
- Resolved distributed system failures (NATS) during AWS Spot Instance auto-scaling events by re-architecting server replica configurations.
- Conducted comparative database benchmarking (MongoDB vs. Couchbase) to drive architectural technology selection.

Projects

Lilac

- Designed and implemented a cross-cloud IaC lifting framework that automatically reconstructs Terraform configurations from existing deployments (“brownfield” infrastructure) across Azure, Google Cloud, and AWS.
- Developed a **pipeline** that integrates Large Language Models (LLMs) with symbolic verification to learn resource–dependency mappings from cloud APIs.
- Conducted large-scale evaluation on real cloud environments, achieving higher accuracy and coverage than existing tools (e.g., Terraformer) while maintaining correctness.

VizThinker

 VizThinker

- Created an innovative graph-based interface for interacting with LLMs, reimagining traditional linear chatboxes into a visual conversation graph.
- Implemented branching and node-based history navigation to support complex idea exploration.
- Deployed on Google Cloud Platform with a modern stack of Node.js, React, and Python.

File-Translator

 file-translator

- Developed a tool using LLM APIs to translate English PDF documents into Traditional Chinese, strictly preserving original formatting and layout.

OS-Nachos

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- Implemented core OS concepts including system calls, multiprogramming, virtual memory, and file systems in C++.

MIPS-CPU-Architecture

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- Constructed a MIPS CPU architecture from the ground up using Verilog, implementing instruction fetching, decoding, and execution.

Advanced Compiler

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- Implemented Data Dependency Analysis using LLVM IR to identify flow, anti, and output dependencies in C programs.
- Developed a Pointer Analysis data flow analyzer for C programs, handling scalar variables and two levels of pointers.